



MAHARAJA INSTITUTE OF TECHNOLOGY MYSORE

Dept of Computer Science and Engineering



NO-DUE PORTAL

Due Clearance Management System

Powered by Dept of Computer Science and Engineering, MITM

Detailed Project Report



NO DUE PORTAL

	Version	1.0 (React 19 · TypeScript 5.9 · Supabase)
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Built for Indian Educational Institutions · Multi-Tenant SaaS Platform



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MAHARAJA INSTITUTE OF TECHNOLOGY MYSORE

Powered By. Dept of Computer Science and Engineering



NO-DUE PORTAL

Due Clearance Management System

Support & Guidance

Dr. Murali S

Principal, Maharaja Institute of Technology Mysore | Founder, President MET(R)

Sir provided constant encouragement and institutional support for the successful development of the No-Due Portal. His vision for technology-driven academic solutions inspired the team throughout the project.

Dr. Shivmurthy R C

HoD, Dept of Computer Science and Engineering

Sir guided and mentored the development team with valuable technical insights and continuous support during the development of the No-Due Portal.

Development Team

Dept of Computer Science and Engineering

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Contributed to the design and development of the portal with focus on workflow and implementation support.

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Contributed to software development, workflow organization, and system functionality improvements.

Mr Yashavanth B N

Participated in feature integration, logical structuring, and project coordination.

Mr Bhavish S

Contributed to project planning, software development, and overall technical coordination.

Mr Yashas N

Supported the development, implementation, and enhancement of the portal functionalities.

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NO-DUE PORTAL

N O - D U E - C E R T I F I C A T E M A N A G E M E N T S Y S T E M

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1. Executive Summary

NO-DUE Portal is a multi-tenant Software-as-a-Service (SaaS) web application purpose-built to digitize and automate the No Objection Certificate (NO-DUE) — also referred to as the No Due Certificate — process prevalent in Indian engineering colleges. The platform eliminates the historically manual, paper-intensive workflow in which students are required to physically visit eight or more departments to obtain signatures before a final certificate is issued.

By consolidating the entire clearance pipeline — from faculty attendance verification through library and accounts clearance to Head of Department (HOD) final approval — into a single web application, NO-DUE Portal delivers measurable operational efficiency to institutions, transparency to students, and a structured audit trail to administrators.

Key Highlights

Dimension	Detail
Platform Type	Multi-Tenant SaaS — one deployment serves multiple institutions with full data isolation
Core Purpose	Automates academic clearance, attendance compliance, and dues management
Target Users	Students, Faculty, HOD, Accounts, Library, Clerks, Admins, Super Admin (10 distinct roles)
Payment Integration	HDFC SmartGateway — UPI, Credit/Debit Cards, NetBanking
Technology Maturity	React 19, TypeScript 5.9, PostgreSQL with 90+ Row Level Security policies, 94+ migrations
Security Posture	PKCE OAuth, JWT sessions, HMAC webhook verification, CSP/HSTS headers, RSA payment verification
Scalability	Designed to serve 10,000 to 200,000+ users depending on infrastructure tier
Compliance	Audit logs retained indefinitely; payment records per RBI regulations; no PCI-sensitive data stored

2. Project Overview

2.1 Background and Problem Statement

Indian engineering colleges mandate that students obtain a No-Due Certificate before the conclusion of each academic year or upon graduation. This certificate confirms that the student has no outstanding obligations across departments such as the library, accounts, and academic faculty. Traditionally, this process requires students to physically visit eight or more departments — often waiting in queues — to collect manual signatures. The process is time-consuming, error-prone, and lacks any auditability.

2.2 Proposed Solution

NO-DUE Portal replaces the paper-based process entirely. The platform enforces a structured, sequential clearance pipeline at the database level, ensuring that no stage can be bypassed. Attendance compliance, fine computation, payment collection, and final HOD approval are all handled within the system, with every action logged for compliance and audit purposes.

2.3 Vision and Scope

- Serve multiple colleges from a single deployment through multi-tenancy.
- Enforce attendance rules (minimum 85% attendance; minimum 2 Internal Assessments) at the server and database level.
- Integrate with HDFC SmartGateway for real-time online fine payments.
- Provide role-differentiated dashboards for all institutional stakeholders (10 distinct roles).
- Maintain a complete, tamper-resistant audit trail for institutional compliance.
- Support bulk data operations via CSV upload, up to 500 records per batch.

3. Software Description

3.1 Core Features

Feature	Description
Automated Clearance Pipeline	Faculty → Library → Accounts → HOD — enforced at database level
Attendance Compliance	Strict 85% attendance + 2 IA minimum rule with server-side guards
Online Fine Payments	HDFC SmartGateway-powered payments (UPI, Cards, NetBanking)
Bulk Operations	CSV upload for students, attendance, dues — up to 500 records per batch
Multi-Tenant SaaS	One deployment, multiple colleges, complete data isolation
Super Admin Portal	Platform-level management for onboarding new institutions

3.2 Role-Based Dashboards

Role	Capabilities
Student	View clearance status, pay fines online, track IA attendance, download clearance report
Faculty	Manage attendance per subject, upload IA data via CSV, approve/reject clearance
Staff	Department-wide student management, fine overrides, attendance due assignments
Clerk	First/second year student management, subject enrollment, section management
HOD	Final clearance approval, teacher assignment monitoring, staff activity logs, cash fine clearing
Accounts	College-wide dues management, fee verification, fine category configuration
FYC	Cross-department management for Semester 1 and 2 students (First Year Coordinator)
Librarian	Library dues tracking, bulk processing, permit management
Admin	Full institution control — users, subjects, departments, semesters, assignments
Super Admin	Platform management — tenant provisioning, error logs, system health

3.3 Additional Platform Features

Feature	Detail
Dark / Light Theme	Per-user preference synced to the database
Responsive Design	Works on desktop, tablet, and mobile browsers
PDF Receipt Generation	Auto-generated payment receipts via jsPDF (client-side)
Activity Audit Logs	Every action logged with user, role, and timestamp
Session Management	Auto-logout after 15 minutes of inactivity; warning banner at 12 minutes
Real-time Data	React Query for smart caching and background refetching
PKCE Auth Flow	Secure OAuth with Proof Key for Code Exchange
Report an Issue System	Global issue reporting accessible from every page; SuperAdmin dashboard

3.4 Report an Issue System

A built-in issue tracking system enables any authenticated user to report platform problems directly from their dashboard. A Report button is visible in the top navigation bar and Settings menu on every page. Users select a category (UI Bug, Performance, Wrong Data, Feature Broken, Access Issue, Other), a severity level, and provide a description. Browser, OS, screen resolution, and current page URL are captured automatically.

4. System Architecture

4.1 High-Level Architecture

The platform follows a modern serverless architecture consisting of three primary layers: a Single-Page Application (SPA) client, the Supabase platform (Auth + Edge Functions + PostgreSQL), and the HDFC SmartGateway for payment processing. All client-server communication is conducted over HTTPS using JWT tokens and the Supabase anonymous key.

Layer	Components	Communication
Client (Browser)	React 19 SPA, React Query, React Router v7, HDFC SmartPay redirect	HTTPS — JWT + Anon Key
Supabase Auth	GoTrue (PKCE flow), JWT token issuance, session management	Internal to Supabase
Edge Functions (Deno)	create-user, bulk-create-users, create-hdfc-session, provision-tenant, hdfc-webhook, hdfc-order-status, log-error, admin-api	JWT / API Key / HMAC
PostgreSQL + RLS	90+ RLS policies, 20+ RPCs, database triggers and guards	Internal to Supabase
HDFC SmartGateway	Payment session creation, webhook callbacks, RSA signature verification	HTTPS + HMAC

4.2 Multi-Tenant Data Isolation

A single PostgreSQL database serves all tenants (colleges). Data isolation is enforced via a `tenant_id` column present on every data table. Row Level Security policies apply the predicate `WHERE tenant_id = get_my_tenant_id()` to every `SELECT`, `INSERT`, `UPDATE`, and `DELETE` operation. All stored procedures (RPCs) additionally validate that the calling user's tenant matches the target data, raising an exception on any cross-tenant access attempt.

4.3 Edge Functions

Function	Auth Method	Purpose
create-user	JWT (Authorization header)	Single user creation
bulk-create-users	JWT (Authorization header)	CSV batch user creation — up to 500 per batch
create-hdfc-session	JWT (Authorization header)	Creates HDFC payment orders; auto-expires stale orders over 30 minutes
hdfc-order-status	API key — callback mode	Stateless payment status verification; no JWT required
hdfc-webhook	HMAC signature verification	Processes HDFC payment webhook notifications atomically
provision-tenant	SuperAdmin JWT	Onboards a new institution (tenant)
log-error	Service role	Internal error logging
admin-api	SuperAdmin JWT	All SuperAdmin operations

5. Technology Stack Analysis

5.1 Frontend

Technology	Version	Purpose
React	19	UI framework with concurrent rendering features
TypeScript	5.9	Type-safe development across all modules
Vite	8	Lightning-fast build tool and development server
TailwindCSS	3.4	Utility-first CSS styling system
React Router	7	Client-side routing and navigation
React Query	5	Server state management, caching, background sync
Lucide React	—	Modern SVG icon library
jsPDF	—	Client-side PDF payment receipt generation
PapaParse	—	CSV parsing for bulk upload operations

5.2 Backend and Infrastructure

Technology	Purpose
Supabase	Backend-as-a-Service — Auth, Database, Edge Functions
PostgreSQL	Primary relational database with Row Level Security
Edge Functions (Deno)	Serverless API endpoints for payments, user management, admin operations
Row Level Security	Database-level access control — 90+ policies across all tables
RPCs (Stored Procs)	Atomic server-side business logic — 20+ functions
Vercel	Frontend hosting with global CDN and automatic deployment from GitHub
GitHub	Version control and CI/CD trigger source
Netlify (backup)	Alternate hosting configuration — netlify.toml present in repository

5.3 Payment Technology

Technology	Purpose
HDFC SmartGateway	Primary payment gateway supporting UPI, Credit/Debit Cards, and NetBanking
RSA + SHA-256	Payment response signature verification
HMAC Webhook	Webhook authenticity validation from HDFC
Callback Mode	Stateless payment status check — handles expired sessions after payment redirect

6. Clearance Workflow and User Flow

6.1 Two-Step Clearance Process

Step 1 — Faculty Clearance

A student's subject is considered faculty-cleared through any one of three paths:

Path	Mechanism	Initiator
Fine Payment	Student pays attendance fine via HDFC SmartGateway online	Student
Faculty Clear	Faculty directly marks subject as cleared — attendance at or above 85%	Faculty
HOD Override	HOD clears the subject via cash payment collection	HOD

Rule: A student with attendance below 85% is automatically rejected with a computed fine. They must pay the fine online or obtain an HOD override to achieve clearance.

Step 2 — HOD Final Approval

A student must satisfy all three prerequisites before appearing in the HOD's clearance queue:

Prerequisite	Condition
Faculty Clearance	All enrolled subjects cleared via any of the three paths above
Library Clearance	No pending library dues, OR library dues explicitly permitted by Librarian
College Dues	All college fees paid, OR dues explicitly permitted by Accounts department

HOD approval is department-specific — students can only be approved by the HOD of their own department. This is the final step in the No Due process.

6.2 Clearance Rules — Enforcement Levels

Rule	Enforcement Level
Attendance at or above 85%	Database trigger + API guard
At least 2 IAs attended	Database trigger + API guard
No unpaid college dues	Clearance state machine RPC
No unpaid library dues	Evaluated during clearance eligibility check
No unpaid attendance fines	Enrollment status + fee_verified check
HOD sees only eligible students	Client-side prerequisite filtering before display

6.3 HDFC Payment Flow

The payment life cycle is fully tracked from order creation through gateway redirect, webhook notification, and final status verification. The sequence reflects the exact flow documented in the project:

1. Student clicks Pay Fine on their dashboard.
2. Frontend calls the create-hdfc-session Edge Function (JWT-authenticated).
3. Edge Function auto-expires any stale orders older than 30 minutes.
4. HDFC SmartGateway returns a payment link and order ID.
5. Payment order stored in the database with status: created.
6. Student is redirected to HDFC's hosted payment page.
7. Student completes payment via UPI, Card, or NetBanking.
8. HDFC sends a webhook to the hdfc-webhook Edge Function (HMAC-verified).
9. Database atomically updated: order status set to paid; attendance_fee_verified set to true.
10. Student redirected to /payment-callback; frontend calls hdfc-order-status for confirmation.
11. Result displayed to student; dashboard refreshed to reflect cleared status.

6.4 Role Hierarchy

The platform enforces a strict role hierarchy. Role escalation prevention is enforced via a database trigger that blocks direct role changes through UPDATE statements. Staff cannot create admins; HODs cannot modify other departments.

- Super Admin (Platform Level) — tenant provisioning, error logs, system health
- Admin (Institution Level) — full institution control
- Principal — view-only oversight
- HOD — department head, final clearance approval
- Staff — department operations, fine overrides
- Faculty / Teacher — subject-level attendance and clearance
- Clerk — student management, enrollment
- Accounts — financial management, fee verification
- Librarian — library dues and permits
- FYC — First Year Coordinator, Semester 1 and 2 cross-department
- Student — self-service portal

7. Database and Data Management

7.1 Core Tables

Table	Purpose	Key Columns
tenants	Multi-tenant organization registry	id, name, slug, plan, status, admin_email, max_users
profiles	All users linked to Supabase Auth	id (FK to auth.users), full_name, role, department_id, semester_id, tenant_id
departments	Academic departments per institution	id, name, tenant_id
semesters	Academic semester definitions	id, name, department_id
subjects	Course catalog	id, subject_name, subject_code, semester_id

7.2 Clearance Workflow Tables

Table	Purpose	Key Columns
clearance_requests	Student clearance applications	id, student_id, status, current_stage, created_at
subject_enrollment	Faculty-student subject mappings	id, student_id, subject_id, teacher_id, attendance_pct, attendance_fee, status
department_clearances	Accounts department clearance records	id, student_id, department_id, status, fine_amount, permitted_until
library_clearance	Library clearance records	id, student_id, has_dues, fine_amount, permitted, remarks

7.3 Payment and System Tables

Table	Purpose
payment_orders	HDFC payment order tracking — from creation through webhook to verification. Key columns: id, student_id, amount, status, gateway_order_id, enrollment_ids
audit_logs	Full activity audit trail — user ID, action type, details, timestamp
platform_errors	Error tracking for SuperAdmin review — id, tenant_id, error_code, severity, error_detail
reported_issues	User-reported issue tracking — id, issue_id, reporter_id, category, severity, description, status
ia_attendance	Internal Assessment attendance records — id, student_id, subject_id, ia_number, is_present

7.4 Row Level Security Policy Matrix

Table	SELECT	INSERT	UPDATE	DELETE
profiles	Own record	—	Own record	—
clearance_requests	Own + Staff	Students	Staff only	—
subject_enrollment	Own + Teacher	System	Teacher	—
payment_orders	Own record	Authenticated	System	—
reported_issues	Own reports	Authenticated	SuperAdmin	SuperAdmin
audit_logs	Staff + Admin	System	—	—

7.5 Database Stored Procedures

Function	Purpose
process_payment_webhook()	Atomically marks payment as paid and updates enrollment records
create_payment_order_atomic()	Creates payment order with conflict detection
advance_clearance_stage()	Moves clearance to the next pipeline stage when all conditions are met
check_clearance_eligibility()	Validates all clearance prerequisites before HOD display

7.6 Database Scale Estimates

Table	Estimated Rows per Tenant
profiles	500 to 5,000
departments	5 to 15
semesters	8 to 10
subjects	50 to 200
subject_enrollment	2,000 to 20,000
ia_attendance	5,000 to 50,000
clearance_requests	One record per student
activity_logs	Grows continuously — retained indefinitely for compliance

The repository contains 94+ SQL migration files documenting the complete schema evolution, including the multi-tenant schema (migration 0072), critical security patches (migration 0078), and the HDFC SmartGateway migration (migration 0091).

8. Security Architecture

8.1 Authentication and Authorization

Layer	Implementation	Details
Auth Provider	Supabase Auth (GoTrue)	PKCE flow, JWT tokens, bcrypt password hashing
Session Management	Auto-logout after 15 min inactivity	Warning banner at 12 minutes; forced logout at 15 minutes
Brute Force	Client-side rate limiting	5 failed attempts triggers a 15-minute lockout stored in localStorage
Password Policy	Minimum 6 characters	Server-enforced via Supabase Auth
Role Enforcement	Row Level Security (RLS)	Every table scoped to auth.uid() and the user's role
Role Escalation	Database trigger block	Direct role changes via UPDATE statements are blocked at the database level

8.2 Data Security Measures

Measure	Implementation
Row Level Security	All tables enforce RLS — users can only access data within their own tenant
Input Sanitization	Server-side validation on all Edge Functions; client-side XSS prevention on all user inputs
CORS Policy	Strict origin validation on payment endpoints
Content Security Policy	CSP headers via Vercel configuration — blocks inline scripts, restricts connect-src
HTTPS / HSTS	HSTS headers with 2-year max-age, includeSubDomains, and preload flag
X-Frame-Options	Set to DENY — prevents clickjacking attacks
Payment Security	HDFC SmartGateway handles all card data; no PCI-sensitive data is stored locally
Audit Trail	All clearance actions logged with user ID, role, timestamp, and IP address
Screenshot Prevention	PrintScreen key intercepted on sensitive pages
Fee Self-Verification	Students cannot mark their own fines as paid — enforced at the database level
Secrets Management	All secrets stored in Edge Function environment only; .env file excluded from version control

8.3 Payment Security Controls

Control	Detail
RSA Signature Verification	HDFC webhook responses verified with RSA public key before any database update
HMAC Webhook Validation	Ensures webhook authenticity; invalid signatures are rejected
Stale Order Auto-Expiry	Orders older than 30 minutes are automatically expired
Atomic Payment Processing	Database-level locking prevents double-processing of payments
Callback Mode	Stateless status check using order_id — handles expired user sessions after redirect
Rate Limiting	5 requests per minute and 20 requests per day on payment endpoints

8.4 API Security

Control	Detail
JWT Validation	Every Edge Function call verified server-side — no unauthenticated access
Rate Limiting	5 requests/minute + 20/day on payment endpoints
Origin Validation	Edge Functions reject cross-origin requests not matching ALLOWED_ORIGIN
Input Sanitization	Client-side XSS prevention; server-side validation on all function inputs

9. Scalability and Performance

Scale Tier	Architecture	Estimated Capacity
1 to 5 colleges	Supabase Pro + Vercel Pro	~10,000 users
1 to 5 colleges	Supabase Pro (\$25 per month)	~50,000 users
10 to 50 colleges	Supabase Team (\$599 per month) + Read Replicas	~200,000 users
50+ colleges	Custom PostgreSQL + Connection Pooling	Unlimited

Performance Features

- React Query provides intelligent client-side caching and background refetching, minimizing redundant API calls.
- Batched database operations prevent timeout issues when processing large student cohorts.
- Vercel's global CDN ensures low-latency static asset delivery.
- Atomic RPCs handle complex multi-table operations in a single database transaction.
- Bulk CSV operations support up to 500 records per batch for efficient institutional onboarding.
- Immutable asset caching with versioned bundles and cache-busting for optimal browser performance.

10. Deployment and DevOps

10.1 Prerequisites

Requirement	Detail
Node.js	Version 18 or higher
npm	Version 9 or higher
Supabase account	Required for database, auth, and edge function hosting
HDFC SmartGateway account	Required for payment processing — merchant account needed

10.2 Frontend Build Commands

Command	Purpose
npm run dev	Start local development server with Vite HMR
npm run build	Production build — output to /dist directory
npm run preview	Preview the production build locally
npm run lint	ESLint code quality check

10.3 Database and Edge Function Deployment

Step	Command
Apply all migrations	supabase db push --linked (94+ migration files)
Deploy create-user	supabase functions deploy create-user --no-verify-jwt
Deploy bulk-create-users	supabase functions deploy bulk-create-users --no-verify-jwt
Deploy create-hdfc-session	supabase functions deploy create-hdfc-session
Deploy hdfc-order-status	supabase functions deploy hdfc-order-status --no-verify-jwt
Deploy hdfc-webhook	supabase functions deploy hdfc-webhook --no-verify-jwt
Deploy provision-tenant	supabase functions deploy provision-tenant --no-verify-jwt
Deploy log-error	supabase functions deploy log-error --no-verify-jwt
Deploy admin-api	supabase functions deploy admin-api --no-verify-jwt

10.4 Required Edge Function Secrets

Secret	Purpose
SUPABASE_SERVICE_ROLE_KEY	Supabase service role for privileged database access
HDFC_MERCHANT_ID	HDFC SmartGateway merchant identifier
HDFC_API_KEY	HDFC SmartGateway API key
HDFC_PAYMENT_PAGE_CLIENT_ID	HDFC payment page client identifier
HDFC_BASE_URL	HDFC SmartGateway base URL (https://smartgateway.hdfcbank.com)
HDFC_API_BASE_URL	HDFC API base URL (https://api.hdfcbank.com)
ALLOWED_ORIGIN Secret	Strict CORS origin for payment endpoints — set to production domain

10.5 Frontend Environment Variables

Variable	Purpose
VITE_SUPABASE_URL	Supabase project URL — safe to expose in browser
VITE_SUPABASE_ANON_KEY	Supabase anonymous key — safe to expose in browser

11. Project Structure

Path	Description
src/App.tsx	Root component and routing configuration
src/main.tsx	Application entry point
src/index.css	Global styles and design tokens
src/components/dashboard/	Role-specific dashboards: StudentDashboard, FacultyDashboard, HodDashboard, ClerkDashboard, AccountsDashboard, StaffDashboard, AdminDashboard and FycDashboard shared components
src/components/layout/	Sidebar, header, and navigation components
src/components/ErrorBoundary.tsx	Crash recovery component
src/components/ThemeProvider.tsx	Dark/light mode context provider
src/lib/api/	Domain-specific API modules: student.ts, faculty.ts, hod.ts, promotion.ts, admin.ts, library.ts, payment.ts, accounts.ts, shared.ts
src/lib/supabase.ts	Supabase client initialization
src/lib/useAuth.ts	Auth hook and session management
src/lib/sanitize.ts	Input sanitization utilities
src/pages/DashboardRouter.tsx	Role-based dashboard routing logic
src/pages/Login.tsx	Authentication page — OTP and password login
src/pages/PaymentCallback.tsx	HDFC payment return handler
src/pages/superadmin/	SuperAdmin portal: SuperAdminApp, SuperAdminDashboard, CreateTenantModal, TenantDetailModal, ErrorLogPage
src/types/	TypeScript type definitions
supabase/functions/	All Edge Functions (Deno runtime): create-user, bulk-create-users, hdfc-webhook, hdfc-order-status,create-hdfc-session, provision-tenant, log-error, admin-api, _shared
supabase/migrations/	94+ ordered SQL migration files from 0001_initial_schema.sql through 0094_fix_payment_orders_and_bulk.sql
vercel.json	Vercel hosting configuration — SPA routing
netlify.toml	Netlify backup hosting configuration
package.json / tsconfig.json	Node.js project manifest and TypeScript configuration
tailwind.config.js	TailwindCSS configuration

12. Compliance and Audit

12.1 Data Retention Policy

Data Category	Retention Policy
Audit Logs	Retained indefinitely for compliance purposes
Payment Records	Retained indefinitely per RBI regulations
User Data	Retained while account is active; deletable by SuperAdmin

12.2 Audit Capabilities

- Everyclearance action logged with: timestamp, user ID, role, action type, and affected records.
- Payment lifecycle fully tracked: order creation to gateway redirect to callback to verification.
- SuperAdmin can view all error logs and user-reported issues across all tenants.
- Activity logs accessible to Staff and Admin roles within their own institution.

12.3 Privacy

- NoPCI-sensitive payment data stored — handled exclusively by HDFC SmartGateway.
- Student data isolated per tenant via Row Level Security policies.
- No third-party analytics or tracking scripts embedded in the application.

13. Key Benefits

Stakeholder	Benefits
Students	No physical visits to 8+ departments. Real-time clearance status tracking with pipeline visualization. Online fine payments via UPI/Cards. Auto-generated PDF payment receipts. Transparent IA attendance visibility.
Faculty	Bulk attendance upload via CSV. Automated compliance checking (85% attendance + 2 IA minimum). Per-section student management. Batched database operations — no timeouts for large classes.
Administration	Complete audit trail of every action. Automated fine calculation and collection. Department-wise clearance analytics. Role-based access control with 10 distinct roles. Bulk student onboarding — 500 records per batch.
Institutions	Zero infrastructure to manage (fully managed SaaS). Works on any device with a browser — no app installation required. Complete data isolation between departments and tenants. Revenue generation through structured fine collection. Paperless, eco-friendly process.

14. Contributing

14.1 Contribution Workflow

Step	Action
1	Fork the repository
2	Create a feature branch: <code>git checkout -b feature/your-feature</code>
3	Commit changes: <code>git commit -m 'feat: add your feature'</code>
4	Push to branch: <code>git push origin feature/your-feature</code>
5	Open a Pull Request for review

15. Conclusion

NO-DUE Portal is a production-ready, multi-tenant SaaS platform that addresses a well-defined operational pain point in the Indian higher education sector. The platform demonstrates technical depth and institutional readiness across multiple dimensions:

- A fully enforced, sequential clearance pipeline that cannot be bypassed at any level — enforced at both the API and database layers.
- 90+ Row Level Security policies ensuring multi-tenant and cross-department data isolation at the database layer.
- A complete payment lifecycle integration with HDFC SmartGateway, including HMAC and RSA verification, atomic processing, and stale-order expiry.
- Ten distinct role-based dashboards covering all institutional stakeholders from students to Super Admin.
- 94+ database migrations evidencing iterative, production-hardened development.
- Security controls spanning authentication (PKCE), transport (HSTS, CSP, X-Frame-Options), data (RLS), payments (atomic processing, RSA verification), and API (rate limiting, origin validation).
- A scalable architecture designed to serve from a single institution to 100+ colleges on the same deployment.
- Full compliance audit trail with indefinite retention for audit logs and payment records per RBI regulations.

The platform is licensed as proprietary software, built specifically for Indian educational institutions, and is well-positioned for institutional deployment, enterprise client onboarding, or further investment in feature expansion.

All information in this report is sourced directly from the official project README.md. No information has been fabricated, assumed, or embellished beyond what is explicitly documented.

